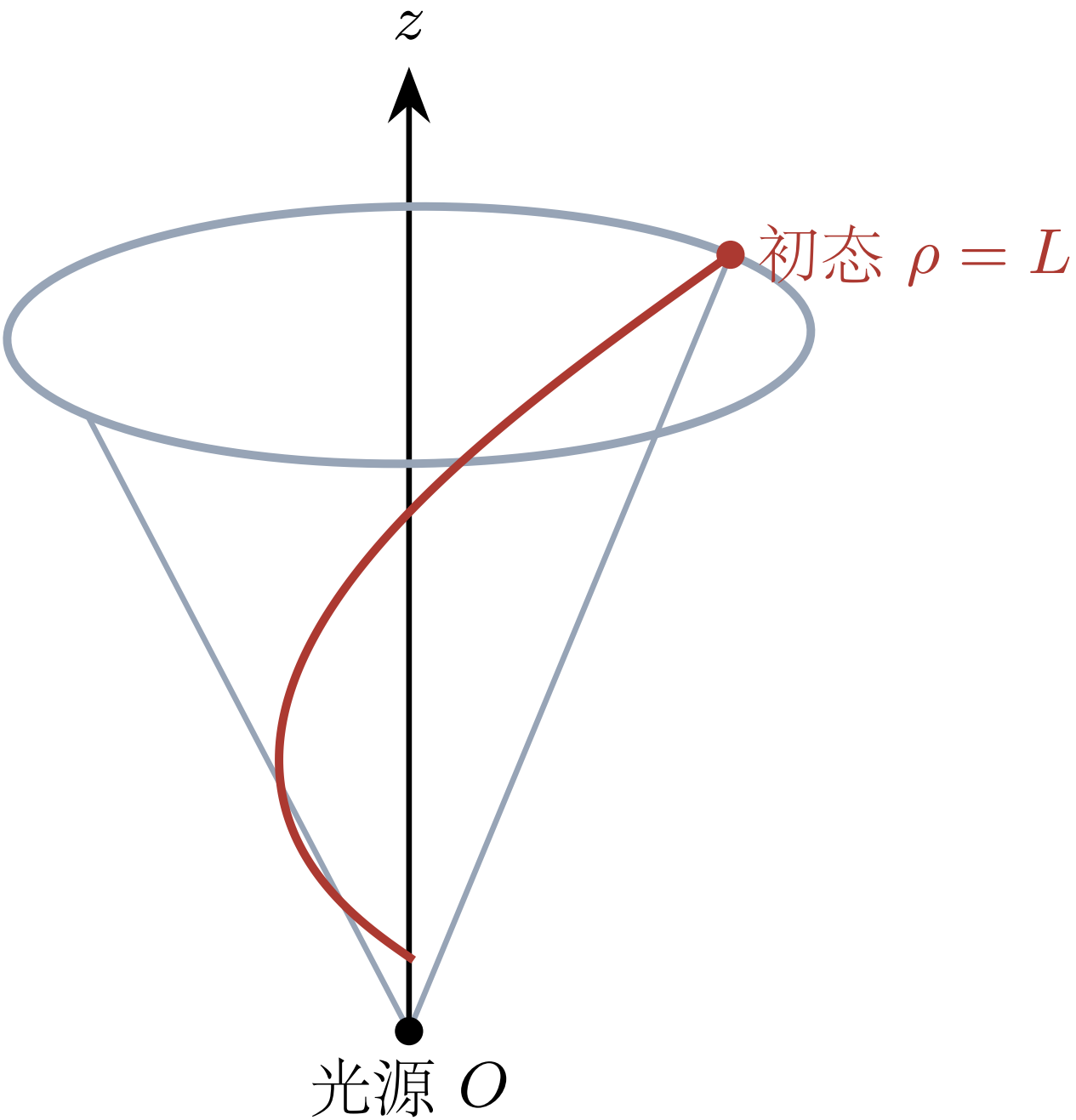
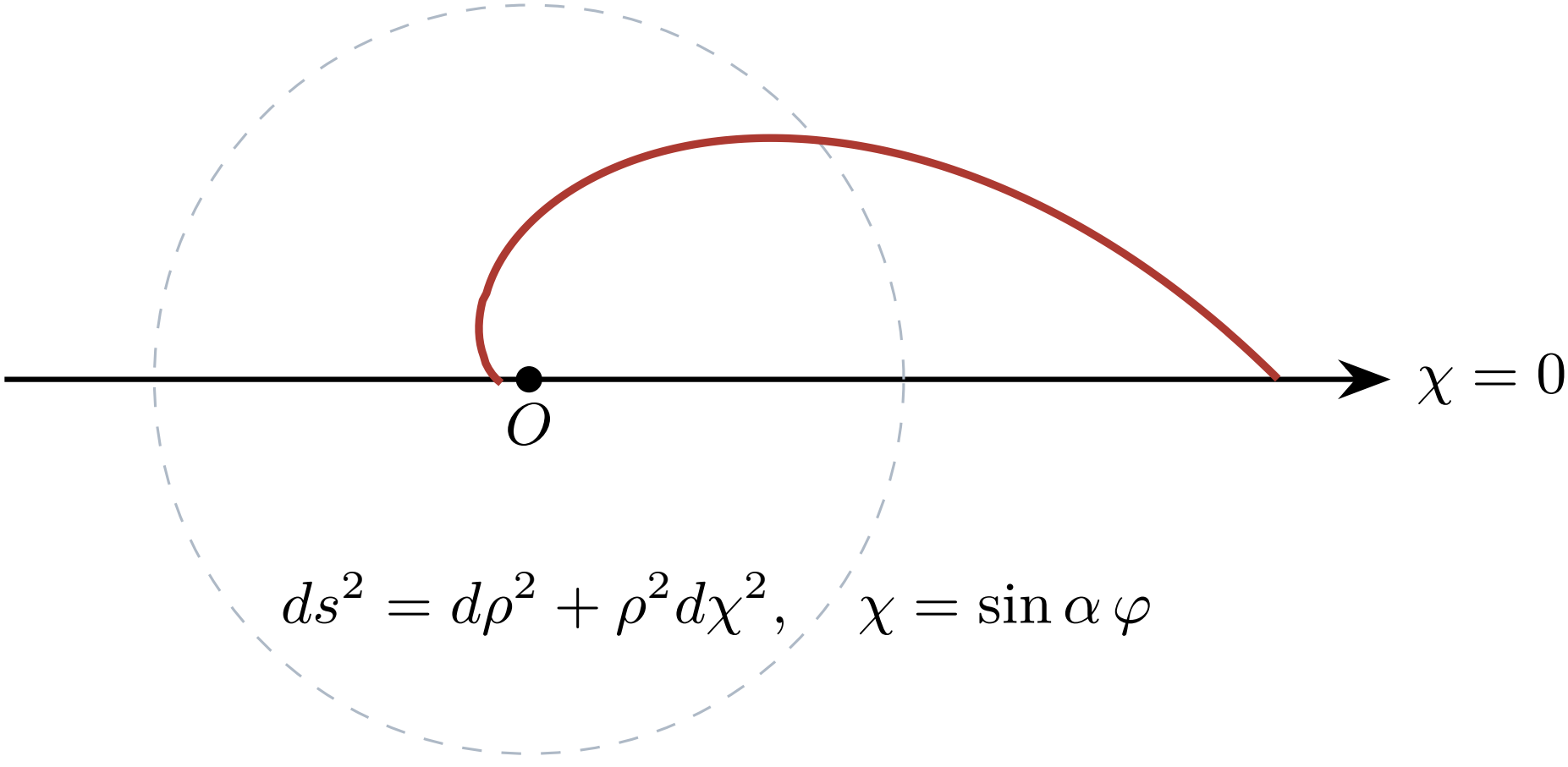
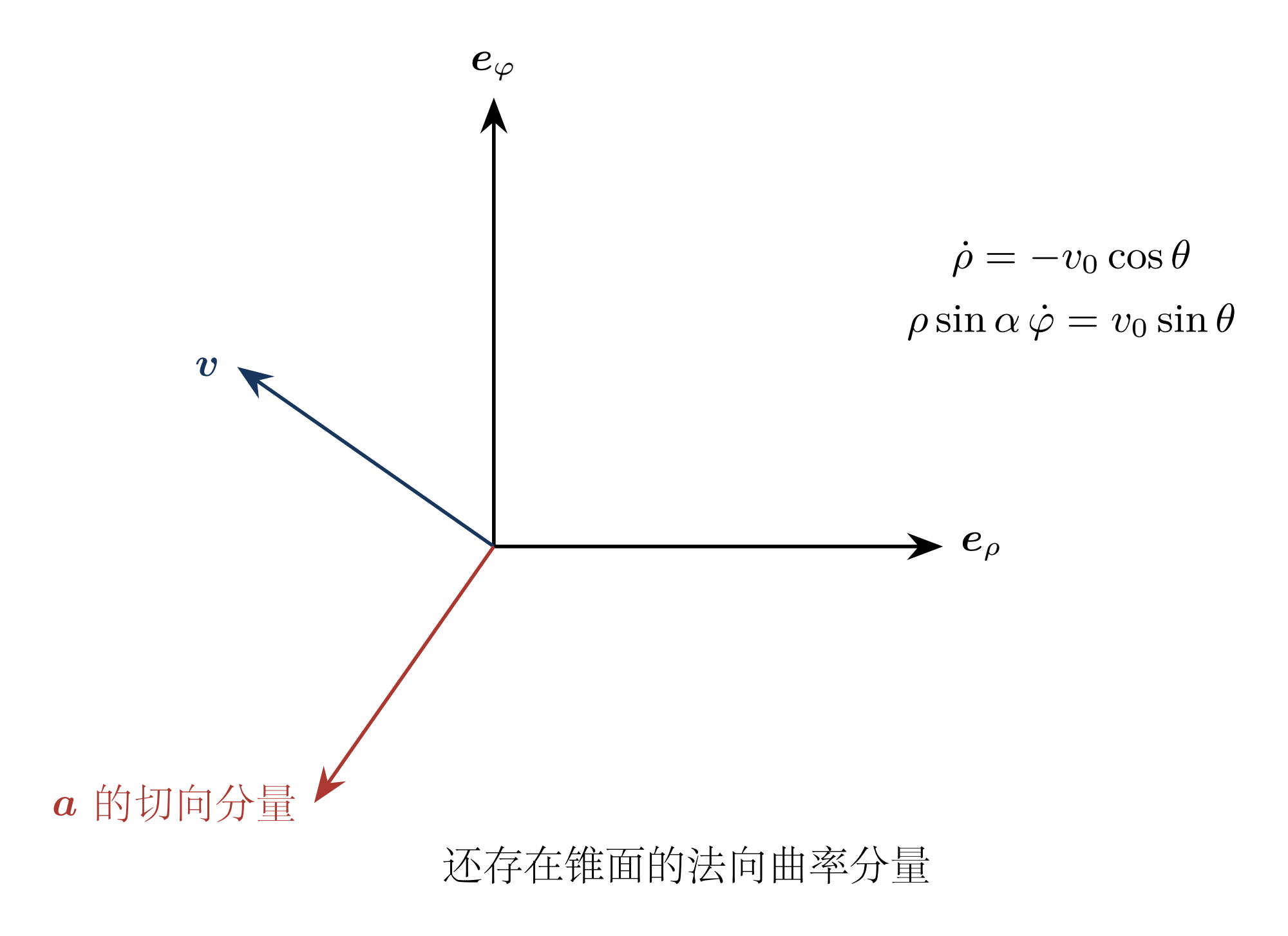

z
初态 $\rho = L$
光源 O

连续展开平面



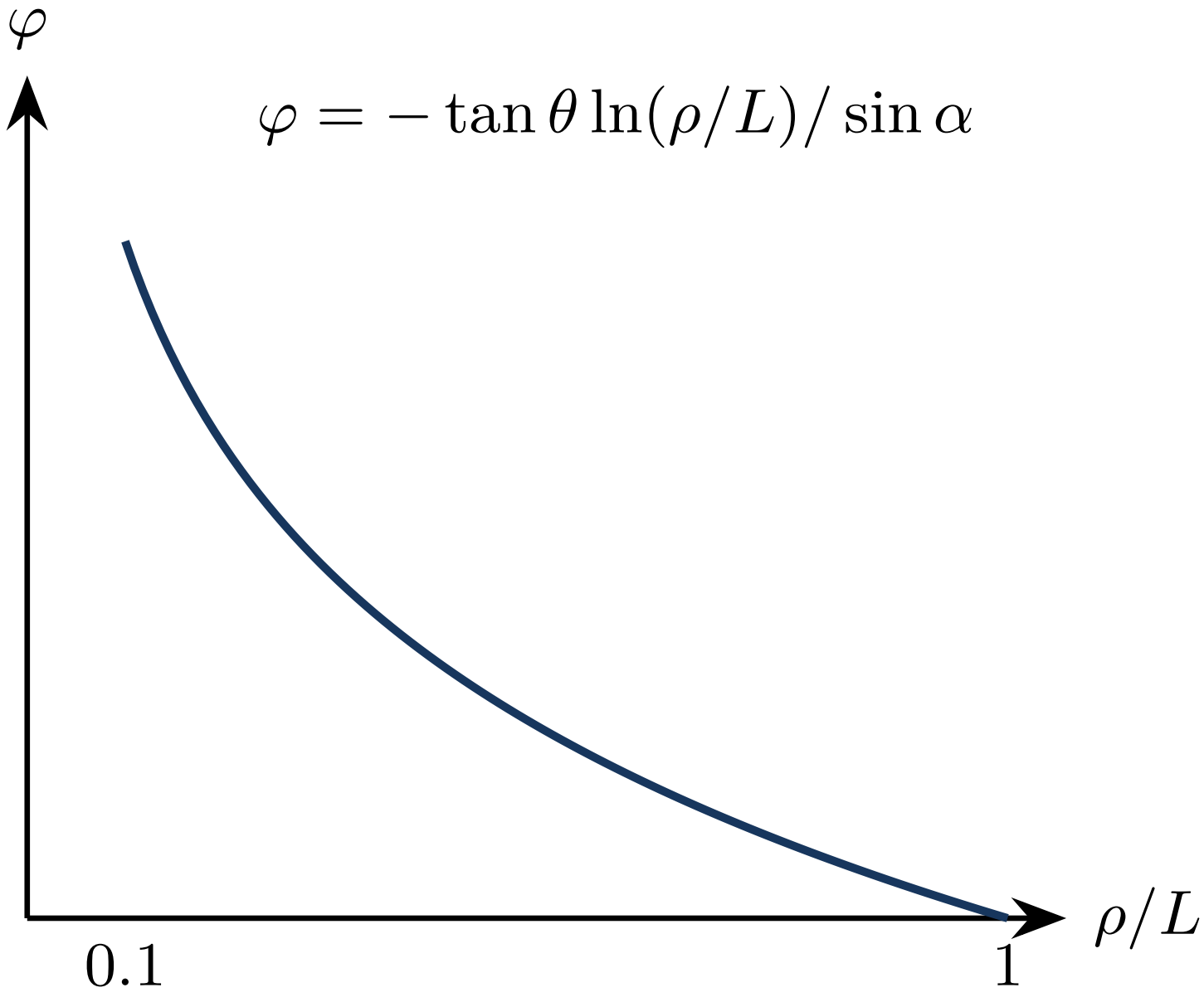
$$ds^2 = d\rho^2 + \rho^2 d\chi^2, \quad \chi = \sin\alpha\,\varphi$$

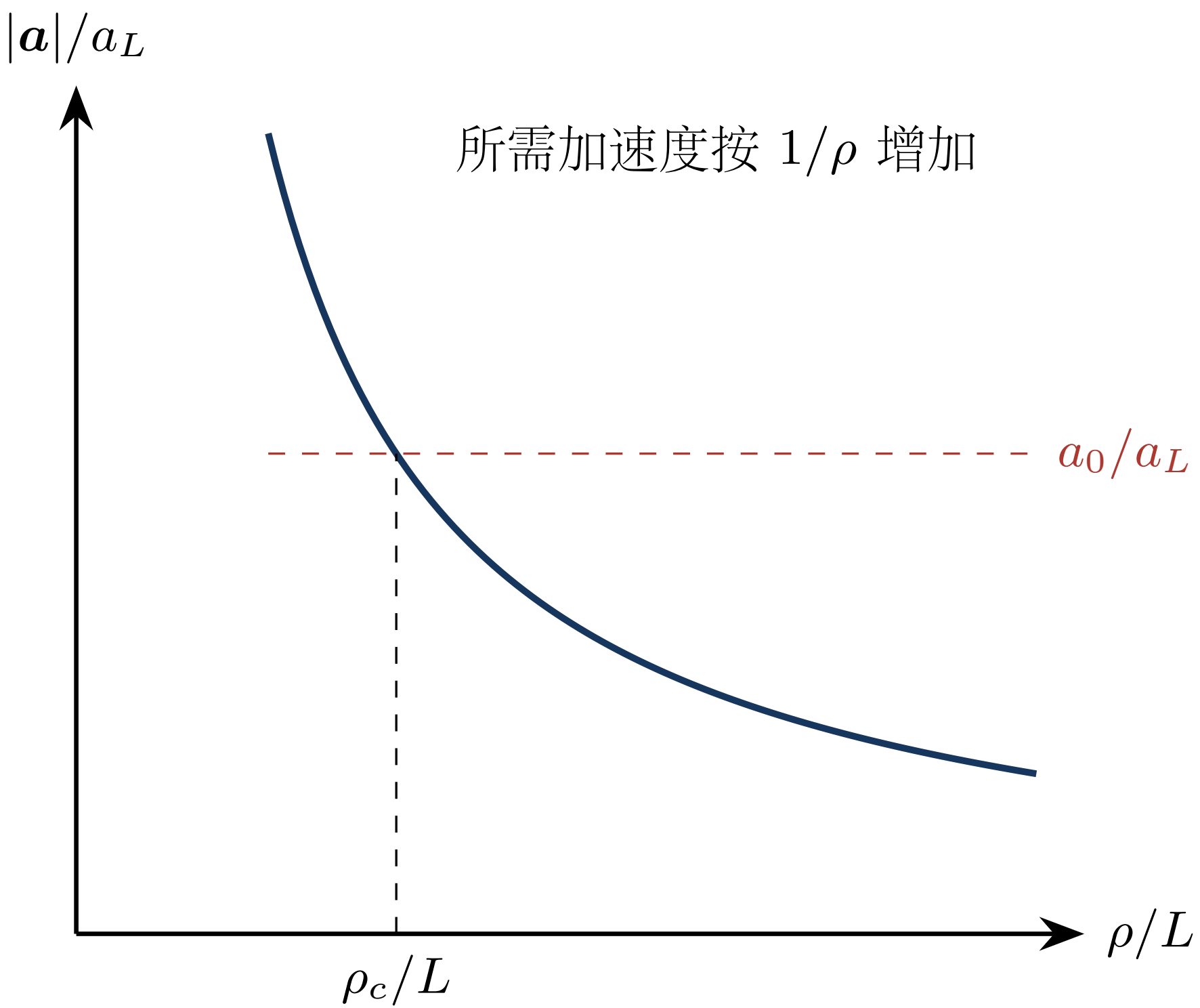


$$\dot{\rho} = -v_0 \cos\theta$$

$$\rho \sin\alpha \, \dot{\varphi} = v_0 \sin\theta$$

还存在锥面的法向曲率分量


φ
$\varphi = -\tan\theta \ln(\rho/L)/\sin\alpha$
ρ/L
0.1
1


$|\boldsymbol{a}|/a_L$
所需加速度按 $1/\rho$ 增加
a_0/a_L
ρ/L
ρ_c/L